

UNIT I: WAVE OPTICS & LASERS

Interference • Newton's Rings • Diffraction • Polarization • Ruby & He-Ne Systems

Interference of Light

$$\Delta = 2\mu t \cos r - \lambda/2$$

$$\text{Stokes Phase Jump: } \pi (\lambda/2)$$

Newton's Rings Air Film

Dark: D

$n^2 = 4n\lambda R \propto \sqrt{n}$ Bright: $D_n^2 = 2(2n-1)\lambda R \propto \sqrt{(2n-1)}$ Diffraction Grating $(a+b)\sin\theta = n\lambda$ Resolving Power: $R = \lambda/d\lambda = nN$ Polarization Brewster: $\mu = \tan i_p$ Malus: $I = I_0 \cos^2\theta$
Laser Engine $A_{21}/B_{21} = 8\pi h\nu^3/c^3$ He-Ne: 632.8 nm

Newton's Rings Formulas Air Film ($2t = D^2/4R$)

- **Dark Ring Diameter:** $D_n = 2\sqrt{(n\lambda R)}$
- **Bright Ring Diameter:**
 $D_n = \sqrt{[2(2n-1)\lambda R]}$
- **Wavelength (λ):** $\lambda = (D_{n+p}^2 - D_n^2) / (4pR)$
- **Liquid Refractive Index:**
 $\mu = (\Delta D^2)_{\text{air}} / (\Delta D^2)_{\text{liq}}$
- **Center Spot:** Always **Dark** in reflected light ($t=0 \Rightarrow \Delta = \lambda/2 \Rightarrow$ destructive).

Fraunhofer Single Slit

 $a \sin\theta$

- **Minima Condition:** $a \sin\theta = \pm m\lambda$ ($m = 1, 2, 3, \dots$)
- **Secondary Maxima:** $\tan \alpha = \alpha$ ($\alpha \approx \pm 1.43\pi, 2.46\pi$)
- **Central Width:** $2\theta = 2\lambda / a$ (Linear = $2f\lambda/a$)
- **Intensity Ratio:** $1 : 0.045 : 0.016 : 0.008$ (>95% in center).
- **Grating:** $(a+b)\sin\theta = n\lambda$
- **Absent Spectra:** $n/m = (a+b)/a$ (If $b=a$, orders 2, 4, 6... absent).

Laser Physics & Types

He-Ne vs Ruby

- **Einstein Relations:** $B_{12} = B_{21}$ | $A_{21}/B_{21} \propto \nu^3$
- **Ruby Laser (3-Level):** Active: $\text{Al}_2\text{O}_3:\text{Cr}^{3+}$ (0.05%), Pump: Xenon flash, $\lambda = 694.3 \text{ nm}$ (Pulsed).
- **He-Ne Laser (4-Level):** He:Ne = 10:1, DC discharge pump, $\text{He } 2^1\text{S} \rightarrow \text{Ne } 3\text{s}$ resonant collision, $\lambda = 632.8 \text{ nm}$ (Continuous red).
- **Retardation:** QWP $t = \lambda / [4(\mu_o - \mu_e)]$ •
HWP $t = \lambda / [2(\mu_o - \mu_e)]$

UNIT II: ELECTROSTATICS, MAGNETOSTATICS & ELECTROMAGNETIC WAVES

Gauss • Poisson/Laplace • Biot-Savart • Ampere-Maxwell • Wave Eq. • Skin Depth • Poynting Vector

Fundamental Law	Differential Equation	Integral Form	Core Physical Interpretation
Gauss's Law (Electrostatics)	$\nabla \cdot \mathbf{E} = \rho / \epsilon_0$	$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}} / \epsilon_0$	Electric charges act as sources/sinks of E-field.
Gauss's Law (Magnetism)	$\nabla \cdot \mathbf{B} = 0$	$\oint \mathbf{B} \cdot d\mathbf{A} = 0$	Magnetic monopoles do not exist; lines form loops.
Faraday's Law of Induction	$\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$	$\oint \mathbf{E} \cdot d\mathbf{l} = -d/dt \int \mathbf{B} \cdot d\mathbf{A}$	Time-varying B generates circulating non-conservative E.
Ampere-Maxwell Law	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$	$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 (I_c + I_d)$	Conduction current + Displacement current create B.

Potentials & Continuity Electro / Magnetostatics

- **Conservative E:** $\nabla \times \mathbf{E} = 0 \Rightarrow \mathbf{E} = -\nabla V$
- **Poisson's Eq:** $\nabla^2 V = -\rho / \epsilon_0$
- **Laplace's Eq:** $\nabla^2 V = 0$ (charge-free region).
- **Magnetic Vector Potential A:** $\mathbf{B} = \nabla \times \mathbf{A}$ with
Coulomb gauge $\nabla \cdot \mathbf{A} = 0 \Rightarrow \nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$.
- **Continuity Equation:** $\nabla \cdot \mathbf{J} + \partial \rho / \partial t = 0$
(Charge conservation).
- **Displacement Current:** $\mathbf{J}_d = \partial \mathbf{D} / \partial t = \epsilon_0 \partial \mathbf{E} / \partial t$.

EM Waves in Free Space

 c & Impedance

- **Wave Equations:**
 $\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \partial^2 \mathbf{E} / \partial t^2$ | $\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \partial^2 \mathbf{B} / \partial t^2$
- **Velocity of Light:**
 $c = 1 / \sqrt{(\mu_0 \epsilon_0)} = 3 \times 10^8 \text{ m/s}$
- **Field Amplitudes:** $E_0 / B_0 = c$
- **Intrinsic Impedance:** $Z_0 = \sqrt{(\mu_0 / \epsilon_0)} \approx 377 \Omega$
- **Transverse Waves:** $\mathbf{k} \cdot \mathbf{E} = 0$ and $\mathbf{k} \cdot \mathbf{B} = 0$ ($\mathbf{E} \perp \mathbf{B} \perp \mathbf{k}$).

Conducting Media & Energy Flow Skin Depth & S

- **Good Conductor ($\sigma \gg \omega\epsilon$):**
 $\alpha = \beta = \sqrt{[\omega\mu\sigma / 2]} = \sqrt{[\pi f \mu \sigma]}$
- **Skin Depth (δ):** Amplitude drops to $1/e$ (36.8%):
 $\delta = 1 / \alpha = \sqrt{[2 / (\omega \mu \sigma)]} = 1 / \sqrt{[\pi f \mu \sigma]}$
- **Poynting Vector:** $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m^2)
- **Poynting Theorem:** Net outflow = Rate of decrease of stored EM energy – Ohmic Joule heat:
 $-\oint \mathbf{S} \cdot d\mathbf{A} = \partial / \partial t \int u_{\text{EM}} dV + \int (\mathbf{J} \cdot \mathbf{E}) dV$

UNIT III: QUANTUM MECHANICS & SCHRÖDINGER EQUATION

de Broglie • Wave Packets • Uncertainty • Operators • 1D Infinite Box • Tunneling • Harmonic Osc.

de Broglie Wavelength

$$\lambda = h / p = h / \sqrt{2mE}$$

$$\text{Electron: } 1.227/\sqrt{V} \text{ nm}$$

Uncertainty Principle

$$\Delta x \cdot \Delta p \geq \hbar/2 \quad | \quad \Delta E \cdot \Delta t \geq \hbar/2$$

$$e^- \text{ Cannot exist in nucleus}$$

Schrödinger Equations

$$\text{TDSE: } i\hbar \partial \psi / \partial t = \hat{H} \psi$$

$$\text{TISE: } \hat{H} \psi = E \psi$$

$$\frac{1}{2} \frac{d^2 \psi}{dx^2} + 2m/\hbar^2 (E - V) \psi = 0 \quad \text{1D Infinite Well (Box)} \quad \psi_n = \sqrt{2/L} \sin(n\pi x/L) \quad E_n = n^2 \hbar^2 / (8mL^2) \quad \text{Harmonic Osc. } E_n = (n+1/2)\hbar\omega \quad E_0 = 1/2 \hbar\omega$$

Wave-Particle Duality

Velocities

$$\bullet \text{ Phase Velocity: } v_p = \omega / k$$

$$\bullet \text{ Group Velocity: } v_g = d\omega / dk$$

$$\bullet \text{ Interrelation: } v_g = v_p - \lambda (dv_p/d\lambda)$$

$$\bullet \text{ Identity: } v_g = v_{\text{particle}} \quad \bullet \quad v_p \cdot v_g = c^2$$

$$\bullet \text{ Operators: } \hat{p} = -i\hbar \nabla \quad | \quad \hat{E} = i\hbar \partial / \partial t$$

$$\bullet \text{ Born Postulate: Probability density}$$

$$P = |\psi|^2 = \psi^* \psi$$

1D Infinite Box (Rigid Well)

Quantization

$$\bullet \text{ Boundary Conditions: } \psi(0) = 0 \text{ and } \psi(L) = 0.$$

$$\bullet \text{ Normalized Eigenfunctions:}$$

$$\psi_n(x) = \sqrt{2/L} \cdot \sin(n\pi x / L)$$

$$\bullet \text{ Quantized Energy: } E_n = n^2 \hbar^2 / (8mL^2)$$

$$\bullet \text{ Energy Ratio: } E_1 : E_2 : E_3 = 1 : 4 : 9 : 16$$

$$\bullet \text{ Zero-Point Energy: } E_1 = \hbar^2 / (8mL^2) \neq 0$$

(required by uncertainty principle!).

Tunneling & Oscillator

Barrier Penetration

$$\bullet \text{ Tunneling Transmission:}$$

$$T \approx e^{-2\alpha a} \quad | \quad \alpha = \sqrt{2m(V_0 - E)} / \hbar$$

$$\bullet \text{ Tunneling Examples: } \alpha\text{-decay, STM microscopes, tunnel diode.}$$

$$\bullet \text{ 1D Harmonic Oscillator: } V(x) = 1/2 m\omega^2 x^2$$

$$E_n = (n + 1/2) \hbar \omega \quad (n = 0, 1, 2, \dots)$$

$$\bullet \text{ Spacing: Equidistant levels } \Delta E = \hbar\omega = h\nu.$$

UNIT IV: CRYSTAL STRUCTURES & FREE ELECTRON THEORY OF METALS

SC/BCC/FCC Master Table • Schottky vs Frenkel • Sommerfeld Theory • Fermi Energy • Richardson Eq.

Crystal Property	Simple Cubic (SC)	Body-Centered Cubic (BCC)	Face-Centered Cubic (FCC)
Effective Atoms (n_{eff})	1 ($8 \times 1/8$)	2 ($8 \times 1/8 + 1$)	4 ($8 \times 1/8 + 6 \times 1/2$)
Coordination Number (CN)	6	8	12
Atomic Radius (r)	$r = a / 2$	$r = (\sqrt{3} / 4) a$	$r = (\sqrt{2} / 4) a = a / (2\sqrt{2})$
Nearest Neighbor (d)	$d = a$	$d = (\sqrt{3} / 2) a$	$d = a / \sqrt{2}$
Atomic Packing Factor (APF)	$\pi / 6 \approx 52.4\%$	$\sqrt{3} \pi / 8 \approx 68.0\%$	$\sqrt{2} \pi / 6 \approx 74.0\%$ (Close-Packed)
Classic Examples	Polonium (α -Po)	Fe, Cr, W, Na, K	Cu, Al, Au, Ag, Pt, Pb

Schottky vs Frenkel Defects

Quick Battle

$$\bullet \text{ Schottky Defect: Equal cation \& anion vacancies. Favored when } r_+ \approx r_- \text{ and high CN. Density decreases! (NaCl, KCl, CsCl).}$$

$$\bullet \text{ Frenkel Defect: Cation displaced into interstitial site. Favored when } r_+ \ll r_- \text{ and low CN. Density unchanged! (AgCl, ZnS).}$$

$$\bullet \text{ Special Exception: AgBr exhibits both defects!}$$

$$\bullet \text{ Schottky Conc: } n_s \approx N e^{-E_s/2kT}$$

$$\bullet \text{ Frenkel Conc: } n_f \approx \sqrt{N(N_i)} e^{-E_f/2kT}$$

Drude vs Sommerfeld

Classical vs Quantum

$$\bullet \text{ Drude Conductivity: } \sigma = n e^2 \tau / m$$

$$\bullet \text{ Wiedemann-Franz Law:}$$

$$K / (\sigma T) = 3/2 (k_B/e)^2 = L$$

$$\bullet \text{ Failures of Classical: Specific heat } C_{v,\text{el}} \text{ is } 100\times \text{ overestimated (predicts } 3/2 R, \text{ observed } 0.01 R); \text{ paramagnetic susceptibility is independent of } T \text{ (not Curie } 1/T).$$

$$\bullet \text{ Sommerfeld Solution: Conduction electrons obey Pauli Exclusion \& Fermi-Dirac statistics. Only electrons within } \sim k_B T \text{ of } E_F \text{ participate!}$$

Fermi Energy & Richardson

Formulas

$$\bullet \text{ Fermi-Dirac Distribution:}$$

$$f(E) = 1 / [e^{(E - E_F)/k_B T} + 1]$$

$$\bullet \text{ At } T = 0 \text{ K: } f(E) = 1 (E < E_F), f(E) = 0 (E > E_F).$$

$$\bullet \text{ At } T > 0 \text{ K: } f(E_F) = 0.5 \text{ (50\% probability).}$$

$$\bullet \text{ Fermi Energy at 0 K:}$$

$$E_F(0) = [\hbar^2 / (2m)] \cdot [3\pi^2 n]^{2/3}$$

$$\bullet \text{ Richardson-Dushman Eq:}$$

$$J = A \cdot T^2 \cdot e^{-\phi / (k_B T)}$$

$$(A \approx 120 \text{ A/cm}^2 \text{K}^2, \phi = \text{Work Function}).$$